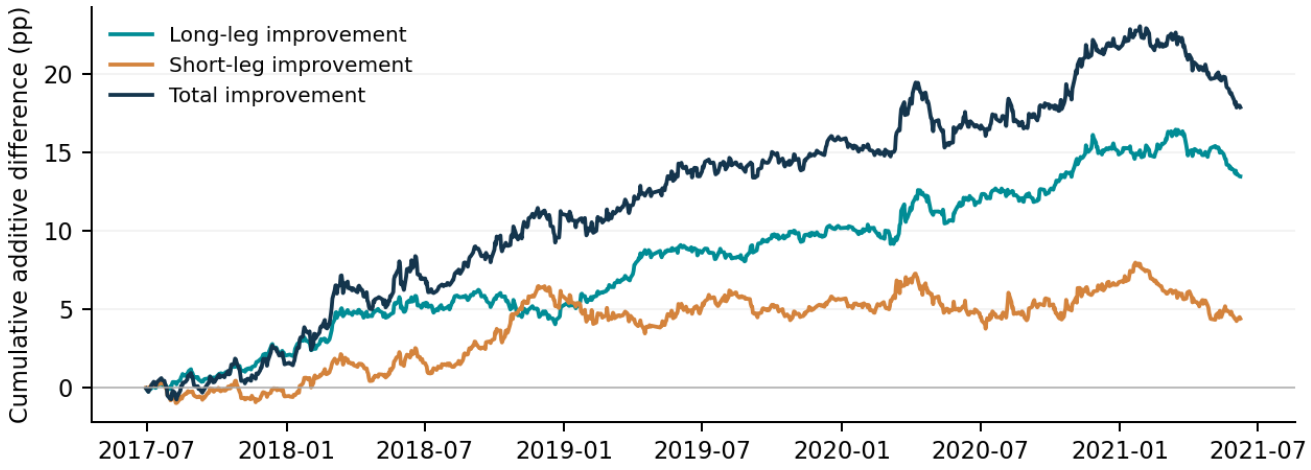
Where did the gain come from?

Matched analysis of 1,853,400 saved out-of-sample predictions per model, over 958 dates and eight folds. Original ReLU vs retrained no-ReLU ensemble. No new training was performed.

The improvement was predominantly on the long side. Long trades contributed 75.3% of the 17.86 percentage-point improvement; shorts contributed 24.7%.

Gross additive contribution	Original ReLU	No ReLU	Change
Long leg	+18.27 pp	+31.72 pp	+13.45 pp
Short leg	-0.49 pp	+3.92 pp	+4.40 pp
Combined spread	+17.78 pp	+35.64 pp	+17.86 pp
Unannualized Sharpe	0.07678	0.14462	+0.06784



Units: each day long weights total +0.5 and short weights total -0.5. Contributions are summed across dates. These are not compounded account returns and exclude trading costs. A positive short contribution means the weighted short basket fell.

Hypothesis assessment

The claim that most of the improvement comes from shorting is not supported. Removing ReLU restored score dispersion, but the biggest incremental benefit came from long selection. Increased forecast negativity alone cannot explain this ranking portfolio.

STUDY 1 | ATTRIBUTION

Improvement versus existing skill

A long portfolio can benefit from a rising universe, while a good short portfolio can still lose money. To separate these, compare each leg with +/-0.5 times the same-day equal-weight return of all eligible stocks. This is a descriptive selection benchmark, not factor-adjusted alpha.

Selection excess, additive	Original ReLU	No ReLU	Improvement
Long minus +0.5 x universe	-0.12 pp	+13.33 pp	+13.45 pp
Short minus -0.5 x universe	+17.90 pp	+22.31 pp	+4.40 pp

The no-ReLU short leg has the larger benchmark-relative selection excess (22.31 pp vs 13.33 pp). However, much of that short-side advantage already existed in the original model. The **incremental change** remains 75.3% long and 24.7% short. Both statements can be true.

Where in time did the improvement occur?

Date group	Long improvement	Short improvement	Total
548 common IC dates	+7.73 pp	+3.20 pp	+10.92 pp
409 added IC dates	+5.73 pp	+1.21 pp	+6.94 pp

The remaining date, 2020-09-29, has constant zero Targets and contributes zero to both portfolios. The added-date group is defined by the original model having a constant score, not by an independently measured difficulty label.

Negative forecasts are not the trading rule

The portfolio shorts the bottom 200 **relative ranks**, even when their forecasts are positive. Negative forecasts account for 56.0% of original short positions and 70.7% of no-ReLU short positions. Adding the same constant to every score on a date would change their signs but leave all ranks and trades unchanged.

The removed activation was between Linear(128,32) and Linear(32,1), not after the final output. The original output could already be negative. Earlier activation replay verified that 97.60% of fold-8 rows had all 32 hidden ReLU outputs zero, leaving only the final bias. This supports a hidden-feature collapse mechanism, not a final-output zero-clipping mechanism.

STUDY 2 | COVERAGE DECOMPOSITION

New dates explain only part of the IC drop

Full-universe daily Rank IC	Original ReLU	No ReLU
All native defined dates	0.01898880 (548 days)	0.00507630 (957 days)
Same 548 dates	0.01898880	0.00955577
Added 409 dates	Undefined	-0.00092553

Coverage rises by 74.64% relative to 548 dates, from 57.20% to 99.90% of all dates (+42.69 percentage points). Native mean IC falls 73.27%. On the same dates alone, it still falls 49.68%.

Arithmetic decomposition

Step	Change in mean IC	Share of total decrease
1. Compare models on the same 548 dates	-0.00943303	67.8%
2. Add the 409 newly defined dates	-0.00447947	32.2%

The decomposition first changes the model on common dates, then expands the date set. Its shares are descriptive and depend on this comparison order; they are not causal percentages.

957-day mean = (548 x 0.00955577 + 409 x -0.00092553) / 957 = 0.00507630.

Conclusion: dilution by added low-IC dates is real, but it explains only about one-third of the observed native-mean decrease under this decomposition. The claim that the decrease is entirely due to coverage is contradicted by the same-date result. A weak IC in these dates does not establish that they are intrinsically hard for every model.

Coverage is a diagnostic, not accuracy

A date is defined only when both scores and Targets vary across stocks. Restoring nonconstant scores removes a failure mode but does not guarantee better ordering. Undefined correlations remain missing; no zeros or random jitter were substituted. All full-universe daily ICs reconcile with the archived analysis.

On the same 548 dates, the 20-date paired block-bootstrap interval for the full-IC change is [-0.01710, -0.00169]. This is an exploratory historical sensitivity interval, without adjustment for selecting this experiment after observing its pilot.

STUDY 2 | TAIL ORDERING

Can the model rank the positions it trades?

For each date, compute Spearman correlation between scores and Targets within that model's predicted top 200, bottom 200, or their union of 400. A positive bottom-200 IC is desirable: lower predicted returns should correspond to lower actual returns.

Rank IC scope	Original mean	Defined dates	No-ReLU mean	Defined dates
All eligible stocks	0.01899	548	0.00508	957
Predicted top 200	Undefined	0	-0.00203	957
Predicted bottom 200	0.03067	548	0.01273	957
Predicted tails combined (400)	0.03399	548	0.01098	957

New finding: the original top-200 scores are identical on all 958 dates. Its long-tail IC is never defined. The model supplies no within-long-basket ordering, and every long cutoff cuts through a score tie. The no-ReLU top-200 baskets contain 199 or 200 distinct scores.

Restrict comparisons to jointly defined dates

Tail scope	Common dates	Original	No ReLU	Change
Bottom 200	548	0.03067	0.02190	-0.00876
Both tails (400)	548	0.03399	0.02043	-0.01356
Top 200	0	Undefined	No paired comparison	N/A

The bottom and combined-tail ICs are also lower without ReLU on common dates. This contradicts a broad claim that every ranking-accuracy measure improved, even though the source-convention portfolio return improved.

Why a low tail IC does not prove luck

Within-tail IC measures the ordering *inside an already selected basket*. It does not directly measure whether the correct stocks entered that basket, the separation between long and short baskets, or the magnitude of gains and losses. Selecting a narrow score range also changes the correlation problem. The no-ReLU top-tail IC is -0.00203, with a 20-date interval [-0.00716, 0.00343]; the short-tail IC is 0.01273, interval [0.00704, 0.01832].

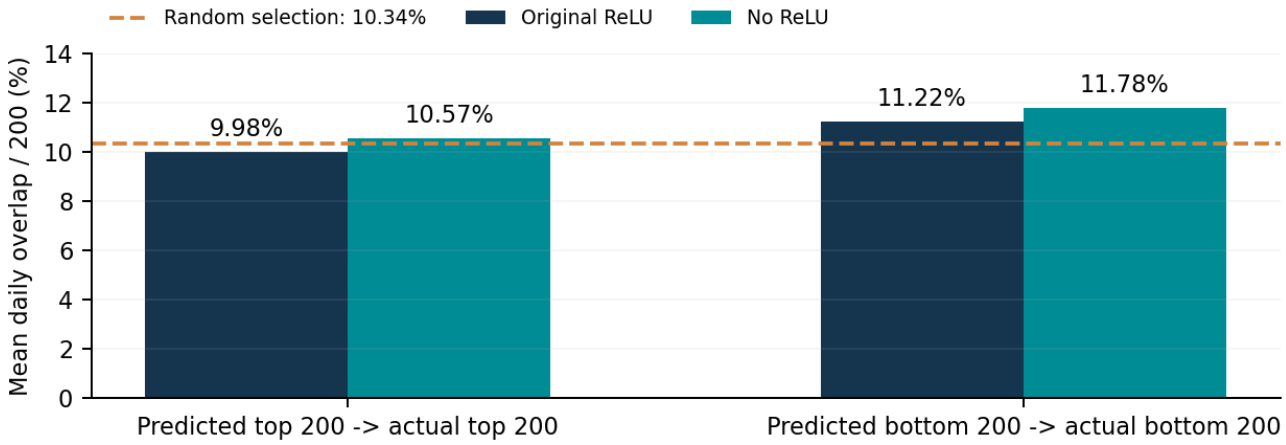
Your proposed rule "tail IC below overall IC implies pure luck" is therefore not a valid falsification test. Tail ordering, membership hits, payoff magnitudes and uncertainty should be examined together.

STUDY 2 | ACTUAL-TAIL RECOVERY

How often are the actual winners and losers found?

Hit rate = overlap between predicted and realized top/bottom 200, divided by 200, then averaged equally over informative dates. With equal basket sizes, this is both precision@200 and recall@200. It does not mean the fraction of positions with a positive trade return.

Membership test	Original	No ReLU	Change	No-ReLU hits / 200
Actual top 200 found	9.98%	10.57%	+0.58 pp	21.13
Actual bottom 200 found	11.22%	11.78%	+0.56 pp	23.56
Uniform random reference	10.34%	10.34%	-	20.68



The random reference averages 200/N for the actual eligible universe on each date. The one date with constant Target is excluded (957 dates remain). At realized-return cutoff ties, allocate remaining top/bottom slots fractionally across equally ranked stocks. This avoids arbitrary stock-code credit. Stable-code tie breaking gives virtually identical percentages.

What the small differences mean

No-ReLU long hits exceed chance by only 0.225 pp; its 20-date bootstrap interval is [-0.165, 0.654] pp and includes zero. Short hits exceed chance by 1.437 pp, interval [1.031, 1.831] pp. Thus evidence for finding actual extreme losers is stronger than evidence for finding actual extreme winners. This is compatible with most *incremental portfolio improvement* coming from the long side.

Relative to the original model, hit-rate changes are +0.584 pp long and +0.561 pp short. Their 20-date paired intervals are [0.074, 1.098] pp and [-0.051, 1.173] pp. Multiple tests and experiment selection are not adjusted.

The Sharpe doubling is sensitive to tie handling

Keep every score and Target fixed, but uniformly randomize stock order **within equal-score groups only**. Use the same position weights and 200 independent complete paths. No jitter is added to scores or to Rank IC.

Tie-order diagnostic	Original ReLU	No ReLU
Long cutoff ambiguous	100.0% of dates	0% of dates
Short cutoff ambiguous	94.68% of dates	0% of dates
Exact expected additive spread	23.92 pp	35.64 pp
Random-tie Sharpe, median	0.14264	0.14462
Random-tie Sharpe, central 95% paths	[0.10384, 0.18008]	[0.14462, 0.14462]

The original Sharpe rises from 0.07678 to a random-tie median of 0.14264, close to 0.14462 without ReLU. Its score-based selection still differs, but the apparent doubling relative to the particular source sorting rule is not robust to this nuisance choice. Original random-tie additive spread has median 23.81 pp and central range [17.60, 30.05] pp. The central 95% of sampled additive differences remain positive: [5.58, 18.04] pp in favor of no ReLU.

These are tie-order sensitivity ranges, not confidence intervals over new markets. The exact expectation averages individual position weights over each tied-score group; it is not the Sharpe of a representative randomized portfolio.

Paired time-block uncertainty under the original sorting rule

No ReLU minus original	Estimate	95% interval, 20-date blocks
Long additive contribution	+13.45 pp	[+3.17, +23.90] pp
Short additive contribution	+4.40 pp	[-6.82, +15.37] pp
Total additive spread	+17.86 pp	[+2.70, +33.93] pp
Unannualized Sharpe	+0.06784	[+0.00520, +0.13642]

2,000 paired circular-block resamples, preserving cross-model date alignment. Five-date results are also in the data bundle. Fixed-rule bootstrap and tie-order sensitivity answer different questions; neither establishes that the gain is purely luck or proves future alpha.

METHODS | INTERPRETATION

What is supported, and what remains open

Proposed explanation	Finding
Most incremental gain comes from shorts	Contradicted: 75.3% long, 24.7% short. Short improvement interval crosses zero.
IC decline is entirely the added dates	Contradicted: common-date IC falls 49.68%; added-date dilution explains 32.2% of the native drop under the stated decomposition.
Removing ReLU restores discrimination	Supported by coverage and score dispersion. This does not imply all accuracy measures improve.
Low tail IC proves pure luck	Invalid criterion. Membership, magnitudes, ties, dependence and out-of-sample replication are needed.
Sharpe nearly doubles robustly	Not supported across tie-order choices. Randomized original ties give median Sharpe close to no ReLU.

Audit and reproducibility

Verified all 167 source manifest entries and exact aligned stock/date/Target rows. Used float32 .npy scores, not rounded printed output. Reconstructed each saved daily spread with maximum absolute discrepancy 9.97e-17, and daily full-universe IC within 1.11e-16. Analysis versions differ from the training environment; daily reconciliation establishes source portfolio equivalence for these saved inputs.

The detail bundle contains the analysis plan and scripts; daily and fold summaries; paired bootstrap results; all selected positions with scores, Targets, source weights and realized-tail memberships; and the prior activation diagnostic. Rerun analyze.py with the original saved JPX_No_ReLU_Ablation_20261018.zip, then supplement.py and build_report.py. No model or optimizer was altered.

Definitions and boundaries

All means of daily IC use defined dates only. Tail comparisons condition on each model's own selected stocks; they compare portfolio behavior, not correlations on identical stock subsets. Full-universe common-date comparisons use identical stock rows. Actual top/bottom membership is a hindsight evaluation label, never a selection input.

Returns use the uploaded JPX Target (next close to following close), source long +0.5 / short -0.5 linear rank weights, and additive aggregation. Costs, slippage, borrow, liquidity and executable fill timing are not reconstructed. This follow-up does not update the prior cost-sensitivity study.

One seed and one historical path; folds share model/Adam state. The experiment was extended after a favorable pilot, and this follow-up is post-hoc. No untouched final test, independent retraining seeds, selection-adjusted inference or factor-neutral attribution has been added. Next validation should fix a tie policy before retraining and test multiple seeds plus a genuinely later holdout.

Source archive SHA256: 78afe7f38f246161725ef79cd3f46cef
2383b6bd9f6876fb591f145718f3807e